Known surveillance: joint likelihood and 1-D profile confidence intervals

Joint 2-D 95% contour | Estimated pair (MLE) | True pair | Separate 1-D 95% profile CIs

(a) p_p = 0.25, p_a = 1.00 | (b) p_p = 0.50, p_a = 1.00 | (c) p_p = 0.75, p_a = 1.00 | (d) p_p = 1.00, p_a = 1.00

(e) p_p = 0.25, p_a = 0.75 | (f) p_p = 0.50, p_a = 0.75 | (g) p_p = 0.75, p_a = 0.75 | (h) p_p = 1.00, p_a = 0.75

(i) p_p = 0.25, p_a = 0.50 | (j) p_p = 0.50, p_a = 0.50 | (k) p_p = 0.75, p_a = 0.50 | (l) p_p = 1.00, p_a = 0.50

(m) p_p = 0.25, p_a = 0.25 | (n) p_p = 0.50, p_a = 0.25 | (o) p_p = 0.75, p_a = 0.25 | (p) p_p = 1.00, p_a = 0.25

Effective reproduction number, R_{eff}

0.60
0.55
0.50
0.45

0.125
0.15
0.175
0.2

Dispersion parameter, k